PHYS 225

Week 14, Lecture 12:
Vector Calculus and Maxwell's Equations

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

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Week 14, Lecture 12

Vector Calculus and Maxwell's Equations

Objectives:

- Review the concepts of line and surface integrals
- Motivate the "fundamental theorems of vector calculus" which relate grad, div, and curl to line and surface integrals
- Rewrite Maxwell's Equations as vector differential equations
- Along the way, learn about the Levi-Civita symbol and the Dirac delta function

One Minute Paper: Which of Maxwell's equations involve a surface integral? Which ones involve a line integral?

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Vector Calculus in Maxwell's Equations

Note before we start:

This is a "math methods" course, not a math course. Much of this material is covered in the (pre- or co-requisite) course MATH 241 (Calculus III). No proofs here, just motivations and examples.

Previous versions of the course focused a lot on the technical details of computing a line or surface integral. My perspective: knowing what they **mean** is much more important for physics. You will compute plenty in MATH 241.

However, you will see things in this lecture you probably didn't see in your math class! Curl is much easier in index notation…

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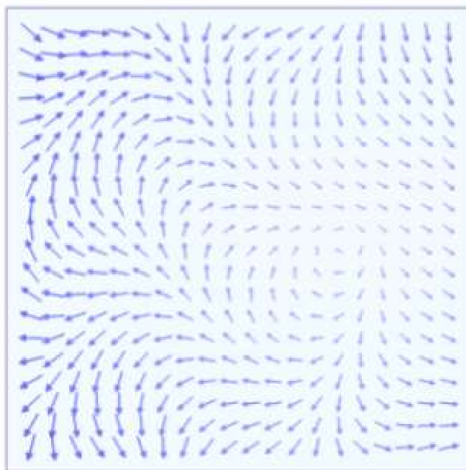
Line integrals

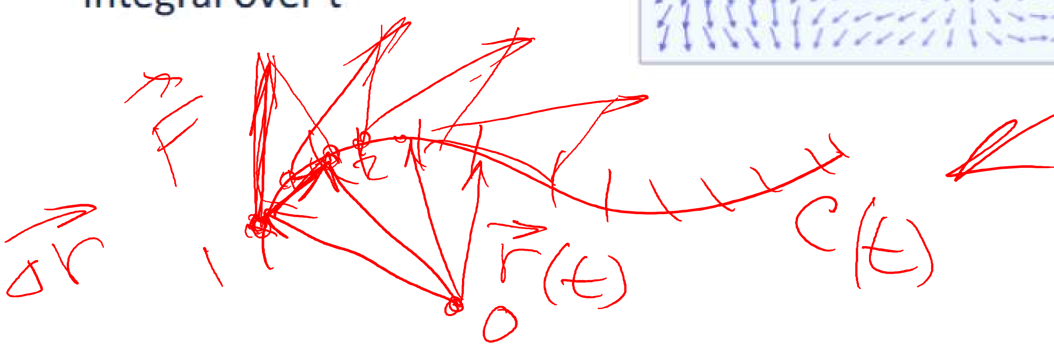
Suppose we have a vector field **F** (e.g. a force), and a curve C (e.g. a path of a particle). We can imagine adding up the component of the vector field parallel to the curve in infinitesimal steps:

Can think of t as the time along the curve: in order to compute, every line integral turns into an ordinary integral over t



$$\lim_{\Delta t \to 0} \sum_{i=1}^{n} \mathbf{F}(\mathbf{r}(t_i)) \cdot \Delta \mathbf{r}_i$$

$$\equiv \int_C \mathbf{F} \cdot d\,\mathbf{r}$$



see line and surface integrals example in Week #13 readings

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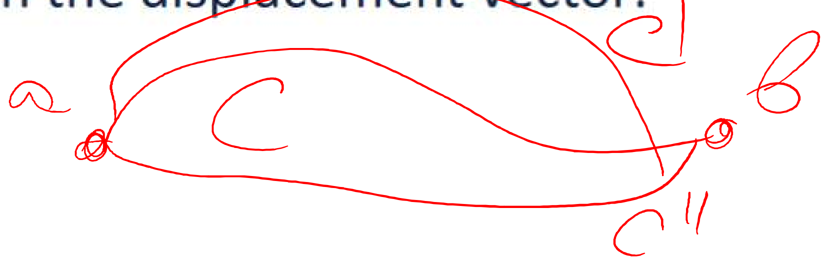
Fundamental theorem for gradients

Fundamental theorem of calculus: $\int_a^b \frac{df}{dx}\,dx = f(b) - f(a)$

This is like a line integral in 1 dimension. The generalization is straightforward: replace the ordinary derivatives with the gradient, and dot with the displacement vector:



$$\int_C \nabla f \cdot d\mathbf{r} = f(b) - f(a)$$

curve C goes from point a to point b

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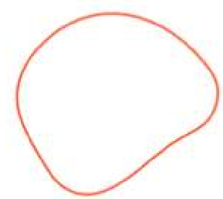
Fundamental theorem for gradients

Two immediate consequences if $\mathbf{F} = \nabla f$ (conservative vector fields):

- the line integral of **F** is path-independent (same answer for all curves C with same endpoints)

- the line integral of **F** around a closed curve is zero

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0$$



closed = no boundary
(= no endpoints)

$$\int_C \nabla f \cdot d\mathbf{r} = f(b) - f(a)$$

curve C goes from
point a to point b

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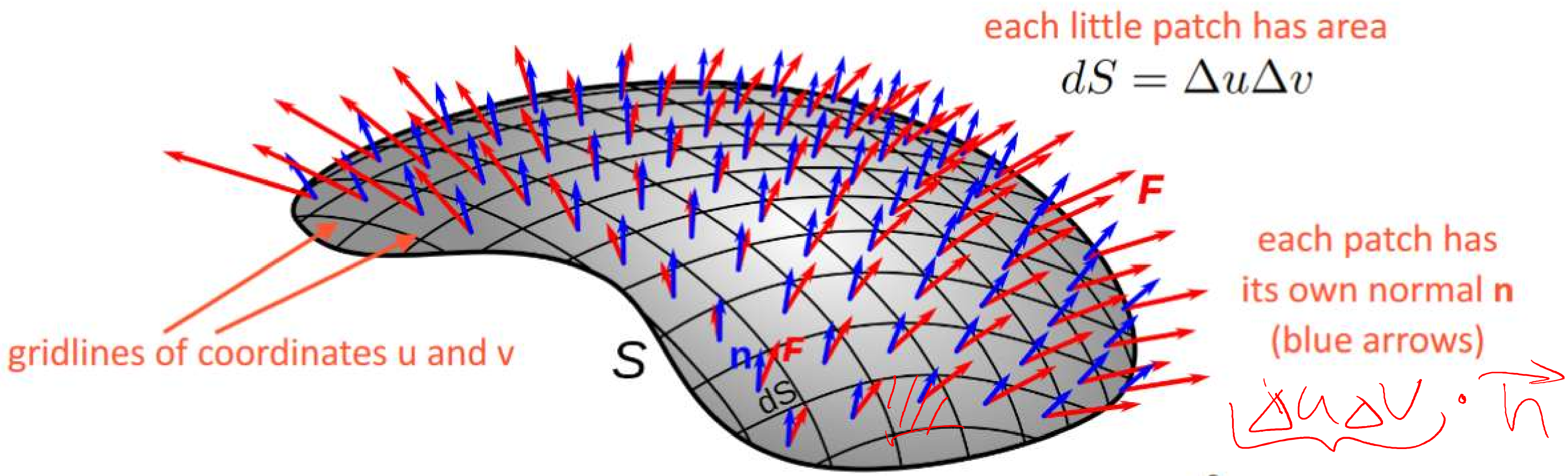
Surface integrals

Suppose we have a vector field **F** (e.g. an electric field), and a surface S.
Can integrate **F** along S, but it's easier in pictures than words:



$$\lim_{\Delta u,\Delta v\to 0}\sum_i \mathbf{F}(u,v)\cdot \mathbf{n}(u,v)_i\,\Delta u\Delta v \equiv \int_S \mathbf{F}\cdot d\mathbf{S}$$

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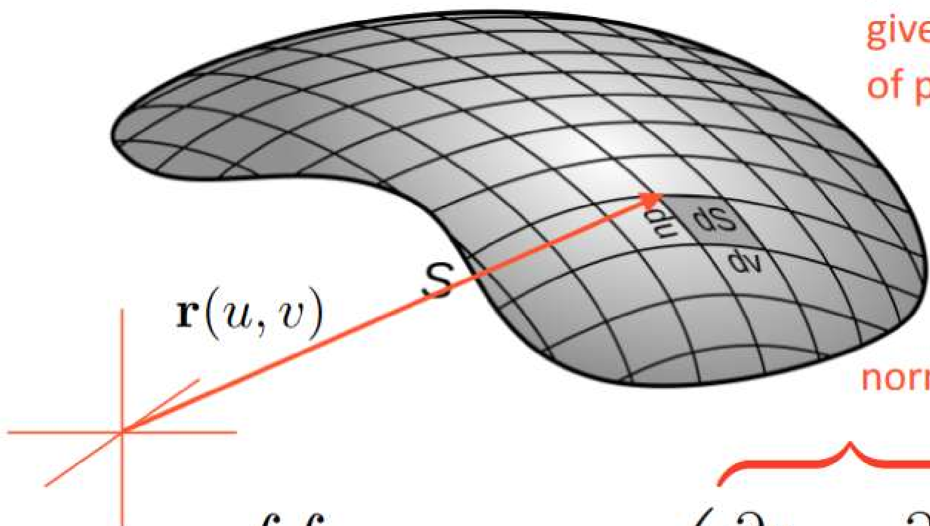
Surface integrals

Just like with line integrals, this turns into an ordinary (double) integral once we parameterize the surface:

positions of points on surface given by **r**(u,v), just like position of point on curve given by **r**(t)



normal vector whose magnitude is the area element dS

$$\int_S \mathbf{F} \cdot d\mathbf{S} = \iint \mathbf{F}(\mathbf{r}(u,v)) \cdot \overbrace{\left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right)} \, du \, dv$$

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More fundamental theorems

All fundamental theorems of calculus have the following form:

$$\int_{\text{region}} \text{derivative of } f = \int_{\text{boundary of region}} f$$

A fancy way to write this is $$\int_R df = \oint_{\partial R} f$$

div, grad, curl as appropriate

boundary of R, which is always **closed** (has no boundary itself)

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$dV = dx\,dy\,dz = $
$= d^3x$

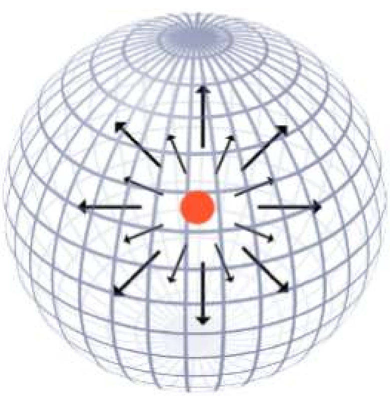
The divergence theorem (Ostrogradsky-Gauss Theorem)

Here is how it works for the divergence:

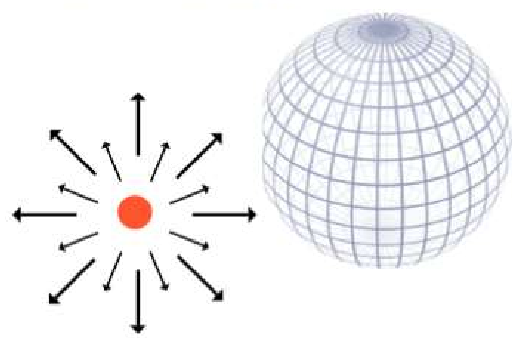
$$\int_V (\nabla \cdot \mathbf{F})\, d^3x = \oint_{\partial V} \mathbf{F} \cdot d\mathbf{S}$$

sometimes written dV

This is why the divergence measures "divergence". Imagine **F** represents velocities of alpha particles emitted from a decaying radioactive source at the origin:



if V contains the origin, surface integral is always positive: things are "diverging" away from origin



otherwise, surface integral vanishes: what comes in must go out

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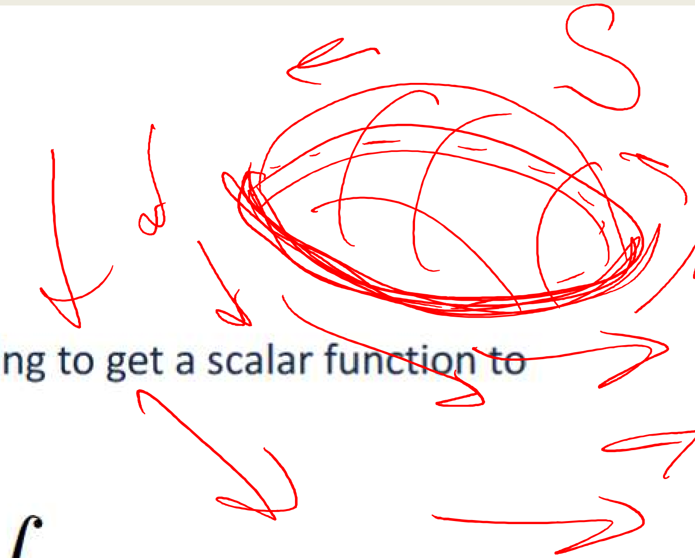
Stokes' Theorem

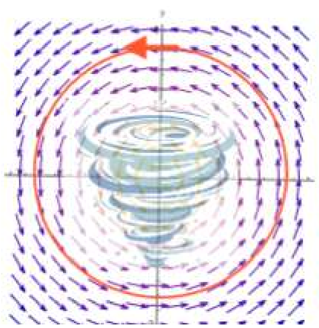


To do this for a curl, need to dot the curl with something to get a scalar function to integrate: the correct choice is a surface integral.

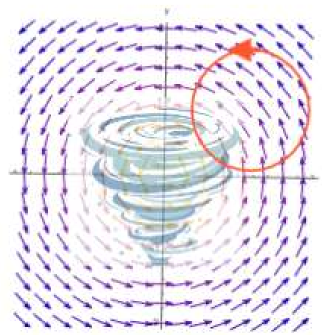
$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$$

circulation of $\vec{F}$ along ∂S

This is why the curl measures "curl", or sometimes "circulation". Imagine **F** represents wind velocities near a tornado which is at the origin:



if S contains the origin, line integral is always positive: things are "circulating" around the origin



line integral vanishes: as many vectors are aligned as anti-aligned

$$\nabla \times \vec{F} = \begin{vmatrix} i & j & k \\ \partial_x & \partial_y & \partial_z \\ F_x & F_y & F_z \end{vmatrix}$$

see line and surface integrals example in Week #13 readings

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Derivative identities from integral theorems

What if **F** itself were the gradient of a scalar function?

$$\int_S [\nabla \times (\nabla f)] \cdot d\mathbf{S} = \oint_{\partial S} \nabla f \cdot d\mathbf{r} = 0$$

Stokes' theorem

Fundamental theorem for conservative vector fields

Since this is true for any f and S whatsoever,

The curl of a gradient is identically zero.

$$\nabla \times (\nabla f) = 0$$

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Derivative identities from index gymnastics

There is a quick way to prove this identity using index notation. First we need a bit of technology:

$$\epsilon^{ijk} = \begin{cases} 1, (ijk) = \text{even permutation of } (1,2,3) \\ -1, (ijk) = \text{odd permutation of } (1,2,3) \\ 0 \text{ otherwise} \end{cases}$$

This is called the (totally antisymmetric) **Levi-Civita symbol**. Swapping two adjacent numbers turns a permutation from even to odd, and vice versa:

$$(1,2,3) \xrightarrow{\text{swap 1 and 2}} (2,1,3) \xrightarrow{\text{swap 1 and 3}} (2,3,1)$$

even	odd	even
$\epsilon^{123} = +1$	$\epsilon^{213} = -1$	$\epsilon^{231} = +1$

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Derivative identities from index gymnastics

The curl in index notation: $\nabla \times \mathbf{F} \longleftrightarrow \epsilon^{ijk}\partial_j F_k$

note the two contractions over j and k

Sanity check: i = 1 is the x-component of the curl:

$\epsilon^{123}\partial_2 F_3 + \epsilon^{132}\partial_3 F_2 +$ (everything else 0) $= \partial_2 F_3 - \partial_3 F_2$

Recall our index notation form for the gradient: $\nabla f \longleftrightarrow \partial_i f$

$$\nabla \times (\nabla f) \longleftrightarrow \epsilon^{ijk}\partial_j(\partial_k f)$$

but this must vanish! Mixed partials commute:
for every $\partial_j \partial_k$ there is a compensating $-\partial_k \partial_j$

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One more derivative identity

Now suppose **F** is the curl of some vector field **A**:

$$\int_V [\nabla \cdot (\nabla \times \mathbf{A})] d^3x = \oint_{\partial V} (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_{\partial\partial V} \mathbf{A} ???$$

"a boundary has no boundary":
there is nothing to integrate over,
so this must vanish.

The divergence of a curl is identically zero.

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

[HW 12: prove this using index notation!]

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A puzzle

Consider the vector field $\mathbf{F} = \dfrac{1}{r^2}\hat{\mathbf{r}}$ in spherical coordinates.

Try to compute its divergence:

$$\nabla \cdot \mathbf{F} = \frac{1}{r^2}\frac{\partial(r^2 F_r)}{\partial r} = \frac{1}{r^2}\frac{\partial}{\partial r}(1) = 0?$$

But if we apply the divergence theorem to a ball B_R of radius R, whose boundary is a sphere S_R of radius R,

$$\int_{B_R} (\nabla \cdot \mathbf{F}) d^3x = \int_{S_R} \frac{1}{r^2}\hat{\mathbf{r}} \cdot d\mathbf{S} = \frac{1}{R^2}\left(4\pi R^2\right) = 4\pi$$

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Dirac delta functions

We should suspicious because **F** blows up at the origin.
In fact, its divergence is proportional to a funny "function" which is **infinite** at the origin and zero everywhere else! This is called a Dirac delta function, and is defined by its integral:

$$\int_V \delta^{(3)}(\mathbf{x})\, d^3x = \begin{cases} 1, & \text{V contains the origin} \\ 0 & \text{otherwise} \end{cases}$$

This solves our puzzle: $\nabla \cdot \mathbf{F} = 4\pi\, \delta^{(3)}(\mathbf{r})$

$$\int_{B_R} (\nabla \cdot \mathbf{F}) d^3x = 4\pi \int_{B_R} \delta^{(3)}(\mathbf{r})\, d^3x = 4\pi$$

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Maxwell's equations in differential form

Start with Gauss's Law:

$$\frac{Q_{\mathrm{encl}}}{\epsilon_0} = \oint \mathbf{E} \cdot d\mathbf{S} = \int (\nabla \cdot \mathbf{E})\, dV$$

Take the integration region to be infinitesimally small, so that the integrand is approximately constant, and look at the infinitesimal charge in that region:

$$(\nabla \cdot \mathbf{E})\, dV = \frac{dQ}{\epsilon_0} \implies \boxed{\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}}$$

charge **density**
$\rho \equiv dQ/dV$

Recalling the field of a point charge, $\mathbf{E} = \dfrac{Q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}}$, we have $\rho = Q\,\delta^{(3)}(\mathbf{r})$

$$\frac{dQ}{dV} = \rho$$

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Maxwell's equations in differential form

Gauss's Law for Magnetism is identical:

$$0 = \oint \mathbf{B} \cdot d\mathbf{S} = \int (\nabla \cdot \mathbf{B})\, dV$$

$$\Longrightarrow \boxed{\nabla \cdot \mathbf{B} = 0}$$ since the volume V can be anything

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Maxwell's equations in differential form

Faraday's Law uses Stokes' Theorem:

$$-\frac{d}{dt}\int \mathbf{B}\cdot d\mathbf{S} = \oint \mathbf{E}\cdot d\mathbf{r} = \int (\nabla \times \mathbf{E})\cdot d\mathbf{S}$$

This holds for any surface, so the integrands must be the same:

$$\boxed{\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}}$$

partial derivative since
B still depends on x,y,z

You'll do the last Maxwell equation (Ampere's law with Maxwell's correction) in discussion!

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Maxwell's equations in differential form

This will be our starting point for the remainder of the course:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

Gauss's Law

$$\nabla \cdot \mathbf{B} = 0$$

Gauss's Law for Magnetism

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Faraday's Law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Ampere's Law (to derive in discussion)

We are almost to our goal of combining E+M with relativity!

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Vector Calculus and Maxwell's Equations

Summary

- Line and surface integrals let us integrate a vector field over a 1-dimensional curve or a 2-dimensional surface
- Generalized fundamental theorems relate integrals of a derivative over a region to an integral over a boundary
- The two vanishing second derivative identities can be derived from Stokes'/divergence/fundamental theorems
- Levi-Civita symbol gives us a compact way to write the curl
- Dirac delta function lets us represent point charges
- Maxwell's equations can be written in differential form using fundamental theorems

One Minute Paper: to apply Stokes' Theorem, do we want to integrate a curl as a line integral or a surface integral?